BIG DATA
CAREER PORTFOLIO

A Six-Month Learning and Project Roadmap

Student Name: Abdul Latheef Nawas
Department: Computer Science and Engineering
College: AMC Engineering College
Programme: B.E. Computer Science and Engineering (CSE–AI&ML), 4th Year
Portfolio Status: Beginner / Learning in Progress

1. Career Objective

My career goal is to build a strong foundation in Big Data and data analytics and gradually prepare for an entry-level role such as Junior Data Analyst, Data Engineer Trainee, or Big Data Intern. As a Computer Science and Engineering student at beginner level, I plan to learn Python, SQL, data handling, data visualization, and Big Data technologies such as Hadoop and Apache Spark. Over the next six months, I aim to complete practical projects, publish clear documentation on GitHub, and work toward a relevant beginner-level certification.

2. Current Profile and Skill Status

Current level: Beginner. This portfolio is a learning plan. Only mark a skill as completed after practising it and being able to demonstrate it.

Area	Current status	Evidence to add
Programming	Beginner; Python practice planned	Small scripts, exercises, and code repository
Databases	SQL fundamentals to learn	Queries using SELECT, WHERE, GROUP BY, JOIN
Data analysis	Planned learning	Cleaned dataset and short findings report
Big Data concepts	Planned learning	Notes explaining volume, velocity, variety, and distributed processing
Visualization	Planned learning	Charts or a simple dashboard
Version control	Git/GitHub setup planned	Public repository with README and project files

3. Career Goal and Target Roles

- Short-term goal: Learn Python and SQL fundamentals and complete one small data analysis project.
- Six-month goal: Build two documented projects and demonstrate basic data-processing and visualization skills.
- Long-term goal: Continue learning cloud data platforms, data pipelines, and scalable processing for a future Big Data or data engineering role.

Possible entry-level roles to explore:

- Junior Data Analyst
- Data Engineer Intern / Trainee
- Big Data Intern
- Business Intelligence Intern

4. Big Data Skills Portfolio

Skill	Why it matters	Learning target / evidence
Python	Automates data processing and analysis.	Write scripts, use functions, and handle CSV files.
SQL	Retrieves and summarizes data from databases.	Practise filtering, aggregation, and joins.
Statistics basics	Helps interpret patterns in data.	Understand mean, median, percentages, and distributions.
Data cleaning	Improves data quality before analysis.	Handle missing values, duplicates, and inconsistent formats.

Data visualization	Communicates findings clearly.	Create labelled charts and explain the key insight.
Hadoop / HDFS	Introduces distributed storage concepts.	Explain the role of HDFS and the components at a beginner level.
Apache Spark / PySpark	Processes larger datasets efficiently.	Run basic transformations and aggregations on a sample dataset.
Git and GitHub	Tracks code and shares project evidence.	Maintain repositories with README, source files, and screenshots.

5. Tools to Learn

Tool	Planned use	Status
Python	Programming and data analysis	To learn / practise
Jupyter Notebook or Google Colab	Run code and document analysis	To set up
Pandas	Clean and analyse tabular data	Planned
SQL with SQLite or MySQL	Query structured data	Planned
Excel	Basic analysis and tables	Planned / update when verified
Power BI	Build a visual dashboard	Optional learning target
Git and GitHub	Version control and project portfolio	To set up
Hadoop and Apache Spark	Learn distributed storage and processing	Planned; begin with concepts and guided labs

Beginner tip: Do not try to master every tool at once. Start with Python, SQL, and data cleaning. Add Hadoop and Spark after the basics are comfortable.

6. Project Portfolio and Evidence

The projects below are proposed projects. They should be marked “Completed” only after the code runs, results are checked, and the repository is published.

Project 1: Student Performance Analysis Using Python and SQL

- Objective: Analyse a sample student dataset to understand marks, attendance, and performance patterns.
- Planned tools: Python, Pandas, SQL, Jupyter Notebook or Google Colab.
- Steps: Obtain a suitable sample or synthetic dataset; inspect and clean it; calculate averages; group results by subject or attendance range; create charts; write a short conclusion.
- Expected evidence: Dataset source, notebook or SQL file, screenshots of results, README, and a short findings summary.

Status: Planned — not yet completed.

GitHub repository link: Repository status: Not published yet; link will be added after implementation.

Project 2: Sales Data Processing and Dashboard

- Objective: Summarize sample sales data by product, month, and region.
- Planned tools: Python/Pandas and either Excel or Power BI.
- Steps: Clean the data, calculate total sales and top-performing products, create charts or a dashboard, and write three useful observations.
- Expected evidence: Cleaned file, analysis notebook or spreadsheet, dashboard screenshots, README, and explanation of the findings.

Status: Planned — not yet completed.

GitHub repository link: Repository status: Not published yet; link will be added after implementation.

Optional Project 3: Introductory Big Data Processing with PySpark

- Objective: Practise reading a sample dataset and calculating grouped summaries with PySpark.
- Planned tools: PySpark in a suitable hosted notebook or local learning environment.
- Evidence: Notebook, explanation of transformations, sample output, and README.

Status: Optional future project; attempt after Python and Pandas basics.

7. GitHub Portfolio Checklist

- ☐ Create a GitHub account and set up a professional profile.
- ☐ Create a separate repository for each project.
- ☐ Add a README describing the problem, dataset source, tools, steps, results, and limitations.
- ☐ Upload code/notebooks and only datasets that are permitted to be shared.
- ☐ Include readable screenshots or sample outputs where useful.
- ☐ Check that code runs and remove passwords, API keys, and personal information.
- ☐ Add the repository links to the final portfolio after they are live.

8. Certification Plan

Completed: Big Data course certificate from Infosys Springboard (issued 18 September 2026). Other certifications below remain planned and will be marked completed only after the course is finished and the certificate is obtained.

Priority	Possible learning option	Evidence / status
1	Python or SQL beginner course (for example, a course from a reputable learning platform)	Planned; record provider, course name, completion date, and certificate link if issued.
2	Introductory data analytics course	Planned; choose one course rather than collecting many incomplete courses.
3	Introductory Hadoop/Spark course or guided lab	Optional; focus on understanding and practical exercises.

Certification record:

Course / certificate	Provider	Status	Date / link
Big Data (Course Completion Certificate)	Infosys Springboard	Completed	18 Sep 2026; verify at verify.onwingspan.com (QR on certificate)
Python / SQL Fundamentals	Provider to be selected	Planned	To be updated after completion

Certificate evidence: Big Data, Infosys Springboard



9. Six-Month Learning Roadmap

Month	Learning focus	Practical activity	Deliverable
1	Python basics: variables, conditions, loops, functions, lists, dictionaries; basic GitHub setup.	Solve small coding exercises and read a CSV file.	Python practice repository and weekly learning notes.
2	SQL fundamentals: SELECT, WHERE, ORDER BY, GROUP BY, aggregate functions, and JOIN; statistics basics.	Query a sample database and answer 10–15 questions.	SQL script with example queries and outputs.
3	Pandas, data cleaning, missing values, duplicates, and charts.	Start Project 1: student performance analysis.	Working notebook, charts, and initial README.
4	Finish Project 1; learn Big Data concepts, distributed storage, and Hadoop/HDFS fundamentals.	Document how traditional processing differs from distributed processing.	Published Project 1 and a short Big Data concepts note.
5	Introductory PySpark; practise reading, filtering, transforming, and aggregating sample data. Learn basic dashboarding if time allows.	Complete a small guided PySpark exercise or start Project 2.	PySpark exercise and/or draft sales analysis.
6	Finish Project 2; review	Test project instructions,	Two documented

	code, documentation, and interview basics; complete one suitable course if possible.	polish GitHub repositories, and prepare a brief project demonstration.	projects, updated portfolio, and certification evidence if completed.

Suggested weekly routine: spend around 5–7 hours each week if your college schedule allows—approximately 2 hours learning concepts, 2–3 hours coding, 1 hour on a project, and 30–60 minutes documenting progress. Adjust the pace to your workload.

10. Monthly Progress Tracker

Month	Planned goal	Status	Evidence / notes
Month 1	Python basics and GitHub setup	Planned	Start Python practice and create GitHub profile.
Month 2	SQL fundamentals and statistics basics	Planned	Save SQL scripts and query outputs.
Month 3	Pandas, data cleaning; start Project 1	Planned	Create notebook and initial charts.
Month 4	Finish Project 1; Big Data and Hadoop concepts	Planned	Publish Project 1 and Big Data notes (Big Data course certificate already earned, Sep 2026).
Month 5	Introductory PySpark; start Project 2	Planned	Save PySpark exercise or sales analysis draft.
Month 6	Finish Project 2; polish portfolio	Planned	Add verified repository links and course evidence.

11. Resume and Interview Readiness

- Add only skills you can explain and demonstrate.
- Describe each project using the problem, tools, work completed, and result.
- Practise explaining a SQL query, a data-cleaning decision, and one chart from a project.
- Keep the resume, GitHub profile, and portfolio consistent.
- Do not list a planned certification as completed or a planned project as professional experience.

12. Conclusion

This portfolio provides a practical starting point for my Big Data career journey. During the next six months, I plan to build foundational skills in Python, SQL, data analysis, GitHub, and introductory Big Data technologies. By completing and documenting small projects, I aim to demonstrate steady progress and prepare for internships or entry-level opportunities. I will update this portfolio with verified project links, results, and certificates as I earn them.

Final Submission Review

- ☐ Fill in student name, college, and academic year.
- ☐ Replace planned statuses with accurate progress updates.
- ☐ Add real GitHub links and screenshots only after projects are completed.
- ☐ Certificates: Big Data certificate attached (Section 8); attach others only when earned.
- ☐ Review spelling, formatting, and all links before submission.

13. Practical Project Implementation Guide

This section turns the roadmap into actions. The project plans below are not represented as completed work. As each task is performed, save the code, output, and a short explanation so the portfolio can be updated with real evidence.

Project 1: Student Performance Analysis — work plan

1. Prepare a small sample CSV with columns such as Student_ID, Subject, Marks, and Attendance_Percentage. Use fictional/sample data only.
2. Load the CSV in Python with Pandas and inspect the number of rows, column names, and missing values.
3. Clean missing or inconsistent values and explain the cleaning choices.
4. Calculate average marks, highest and lowest marks, and subject-wise averages.
5. Create at least two labelled charts and write three observations based on the actual results.
6. Save the notebook, dataset source or synthetic-data note, charts, and README in a GitHub repository.

Completion evidence to attach: notebook or .py file, sample output screenshots, README, and repository URL. Current status: Planned.

Project 2: Sales Data Analysis — work plan

7. Use a permitted sample sales CSV containing Order_Date, Product, Region, Quantity, and Sales_Amount.
8. Check data types, remove duplicate rows where appropriate, and handle missing values.
9. Calculate total sales, sales by month, sales by region, and top products.
10. Build a simple chart set or dashboard in Excel/Power BI and explain three findings.
11. Publish the analysis file and README; mention the dataset source and any limitations.

Completion evidence to attach: cleaned data, analysis file, chart/dashboard screenshots, README, and repository URL. Current status: Planned.

14. Weekly Study Routine

- Monday: Learn one concept and write short notes.
- Tuesday: Practise Python or SQL exercises.
- Wednesday: Repeat exercises without copying a tutorial.
- Thursday: Work on the current project.
- Friday: Test the code and correct errors.
- Weekend: Update GitHub/README and review the week's learning.

Suggested effort: approximately 5–7 hours per week, adjusted around college classes, exams, and project work. Consistency is more important than rushing through every tool.

15. Portfolio Status Declaration

I am beginning my Big Data learning journey as a fourth-year B.E. CSE–AI&ML student. This portfolio records my current beginner status, planned learning activities, proposed projects, and six-month goals. Skills, project links, and certifications will be marked completed only after I have performed the work and can provide evidence.

Student: Abdul Latheef Nawas

College: AMC Engineering College

Programme: B.E. CSE–AI&ML, 4th Year